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Enum

National Instruments

Leader in data acquisition technology with innovative modular instruments and LabVIEW graphical programming software









Dr. James Truchard

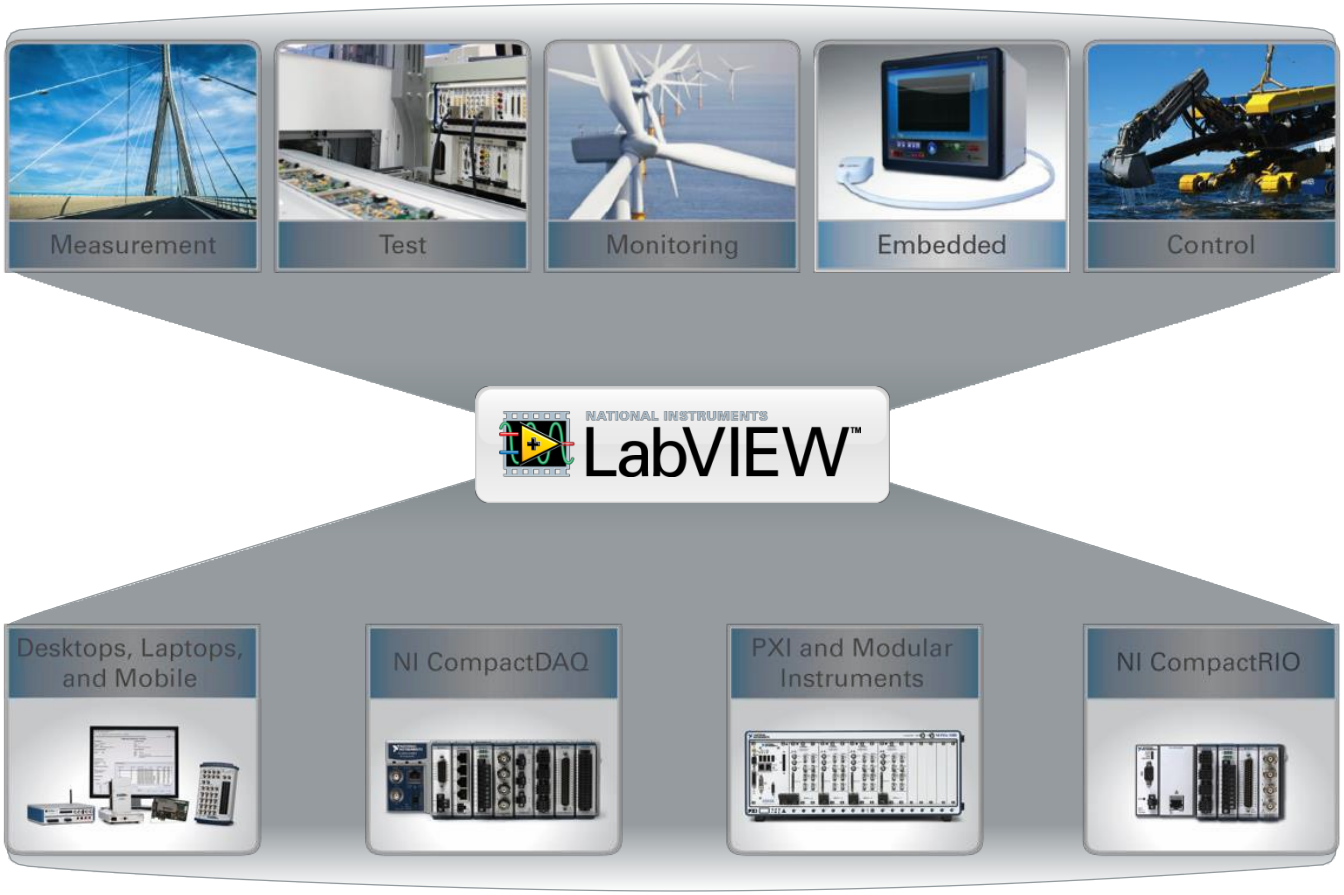
- Corporate headquarters in Austin, TX
- Offices in nearly 50 countries
- 35,000+ companies served annually
- Approx. 7,100 employees
- More than 1,000 products
- 600 Alliance Partners

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Platform-Based Approach



Navigating LabVIEW

TOPICS

A. What Is LabVIEW?
B. Project Explorer
C. Parts of a VI
D. Front Panel
E. Block Diagram
F. Searching for Controls, VIs and Functions
G. Selecting a Tool
H. Dataflow

A. What Is LabVIEW?



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LabVIEW™

Laboratory **V**irtual

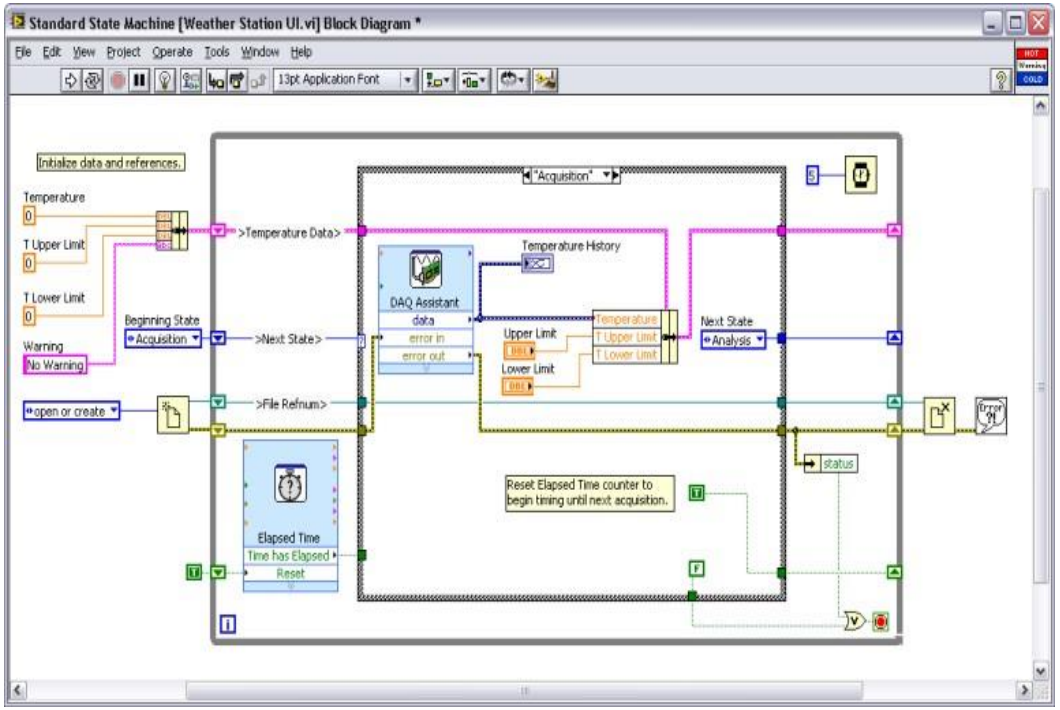
Instrumentation **E**ngineering

Workbench

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What Is LabVIEW?

—A graphical programming environment used to develop sophisticated measurement, test and control systems.



LabVIEW:

- Interfaces with wide variety of hardware
- Scales across different targets and OSs
- Provides built-in analysis libraries

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B. Project Explorer

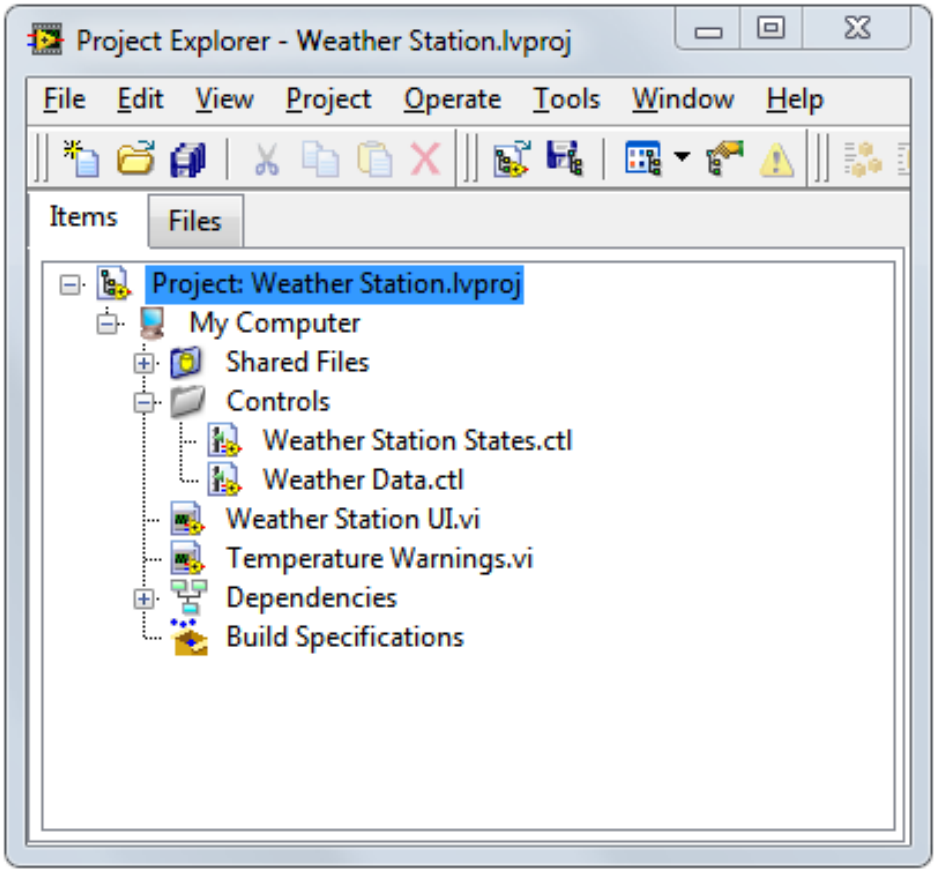
Project Explorer Window

Files Types

Project Folders

Project Explorer

- Find, access, and organize project files
- Prevent, detect, and resolve incorrect links
- Deploy or download files to targets
- Manage code for build options
  - Executables, installers, and zip files
- Integrate with source code control providers



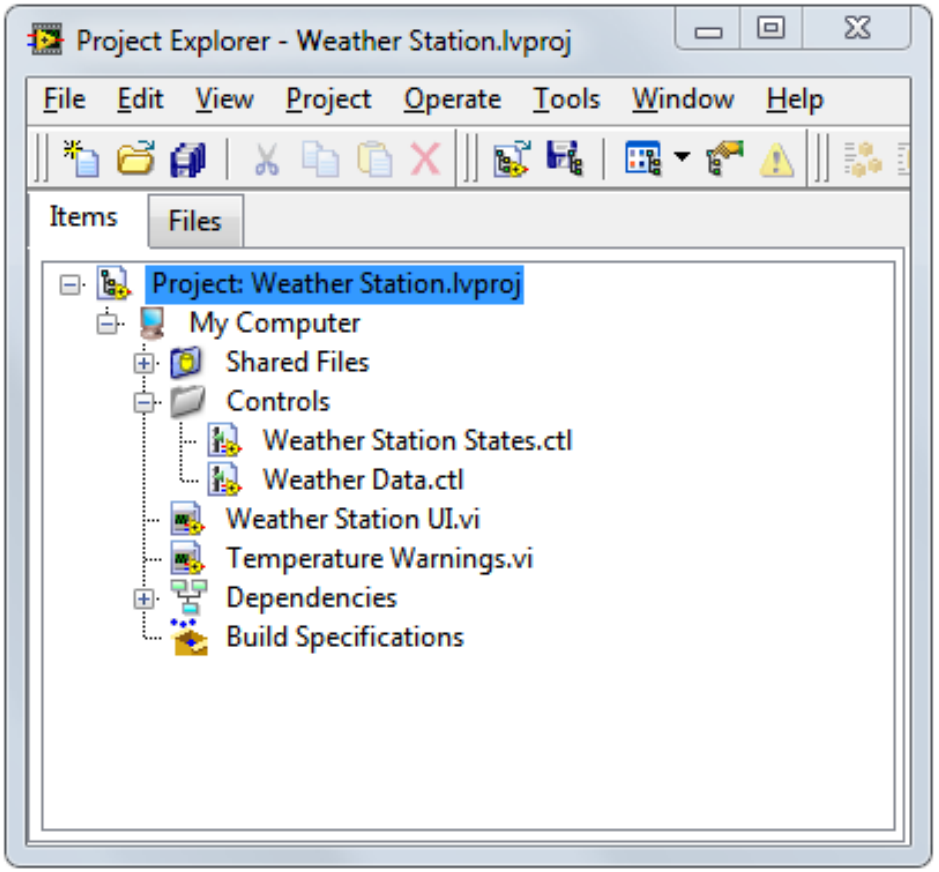
LabVIEW Files

Common LabVIEW file extensions:

LabVIEW project —.lvproj

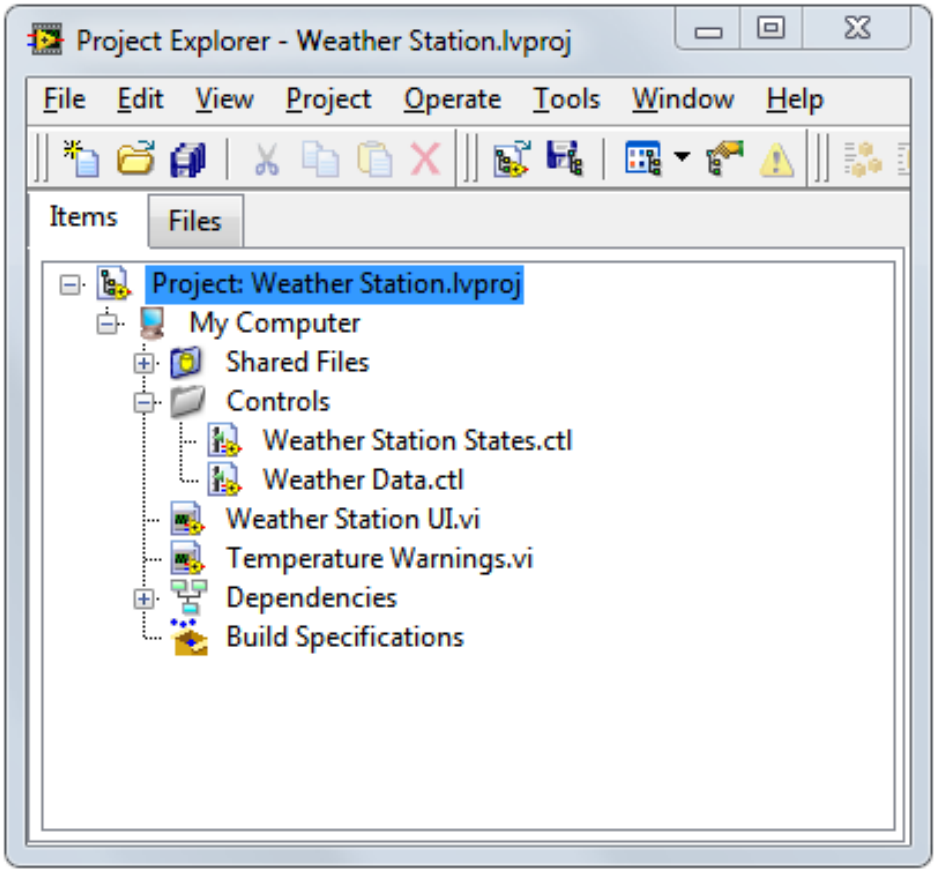
Virtual instrument (VI) — .vi

Custom control — .ctl



Adding Folders to a Project

- Virtual folder
  - Organizes project items and does not represent files on disk
- Auto-populating folder
  - Adds a directory on disk to the project
  - LabVIEW continuously monitors and updates the folder according to changes made in the project and on disk



C. Parts of a VI

Front Panel

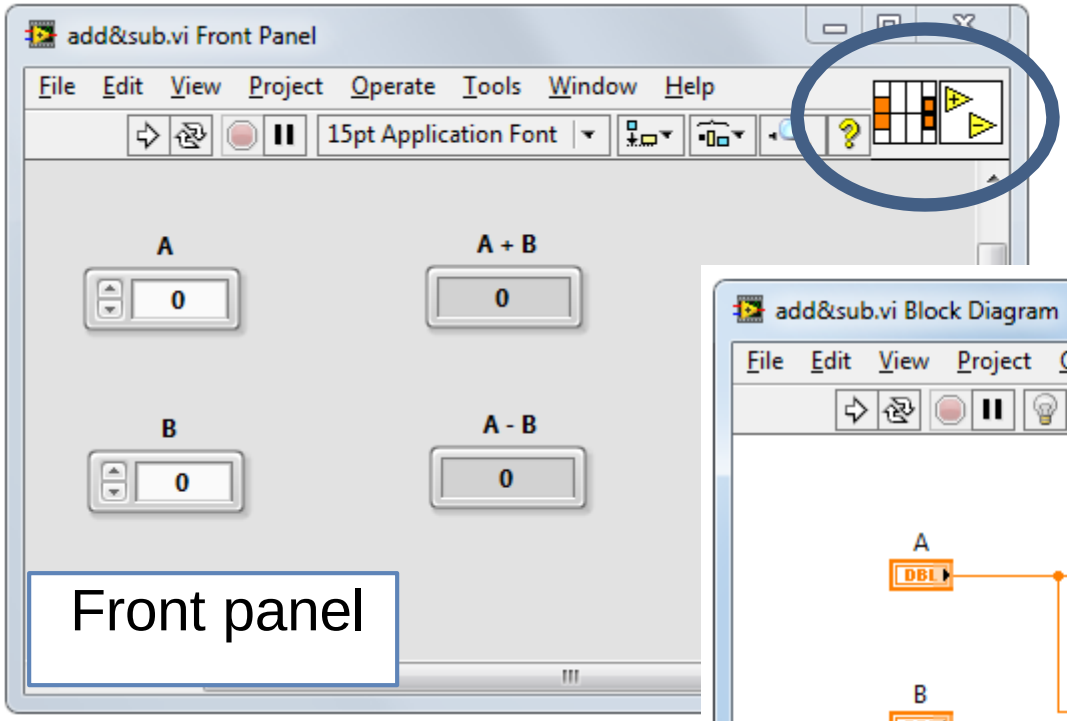
Block Diagram

Icon

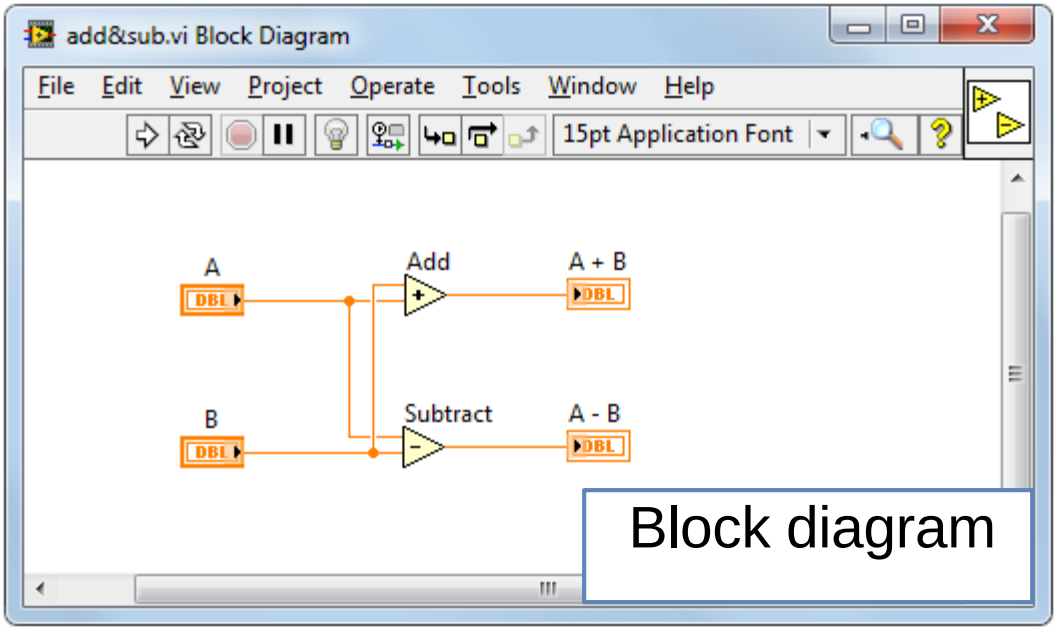
Connector Pane

Parts of a VI

VIs have 3 main components:



Icon/Connector pane



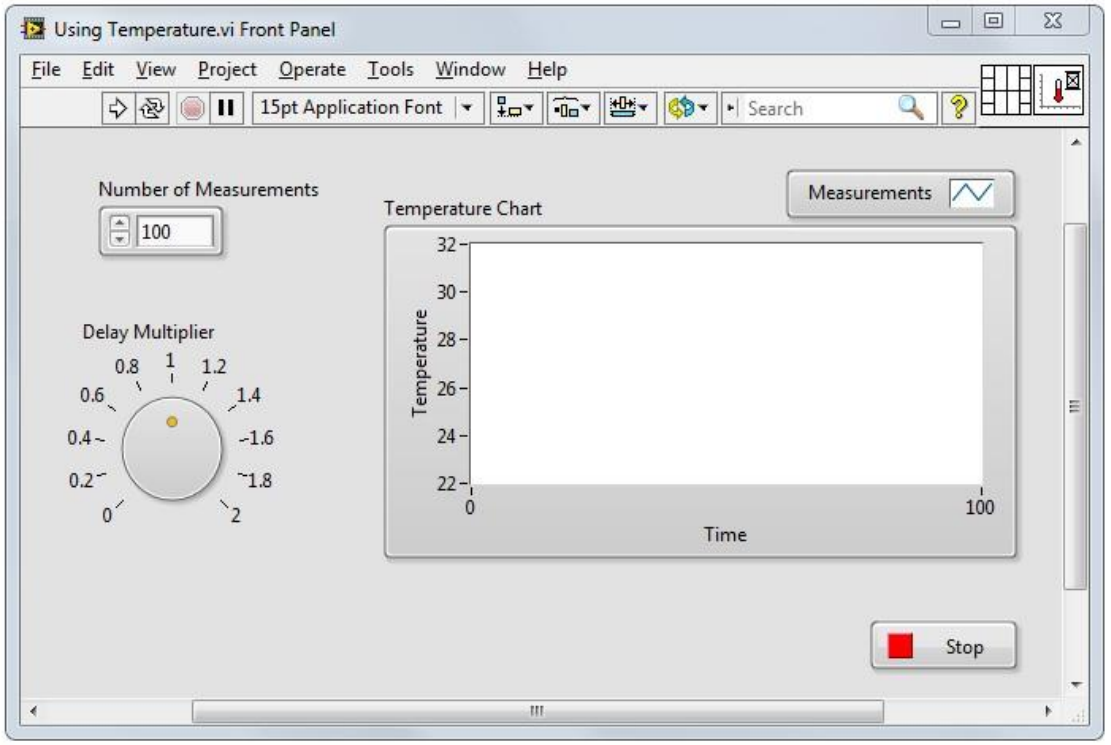
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Parts of a VI – Front Panel

Front Panel – User interface for the VI

You build the front panel with

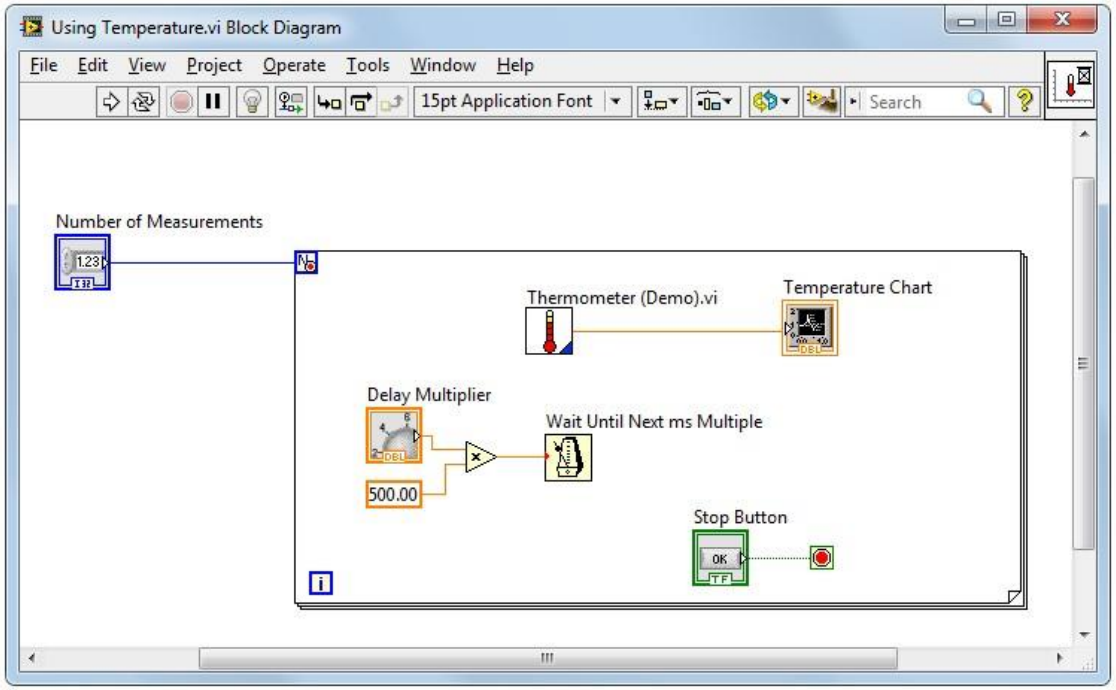
controls (inputs) and indicators (outputs).



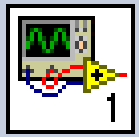
Parts of a VI – Block Diagram

Block Diagram – Contains the graphical source code

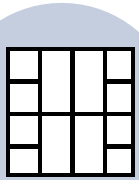
Front panel objects appear as terminals on the block diagram.



Parts of a VI – Icon/Connector Pane



Icon – Graphical representation of a VI



Connector Pane – Map of the inputs and outputs of a VI

Icons and connector panes are necessary to use a VI as a subVI.

- A subVI is a VI that appears on the block diagram of another VI.
- A subVI is similar to a subroutine or function in a text-based programming language.

D. Front Panel

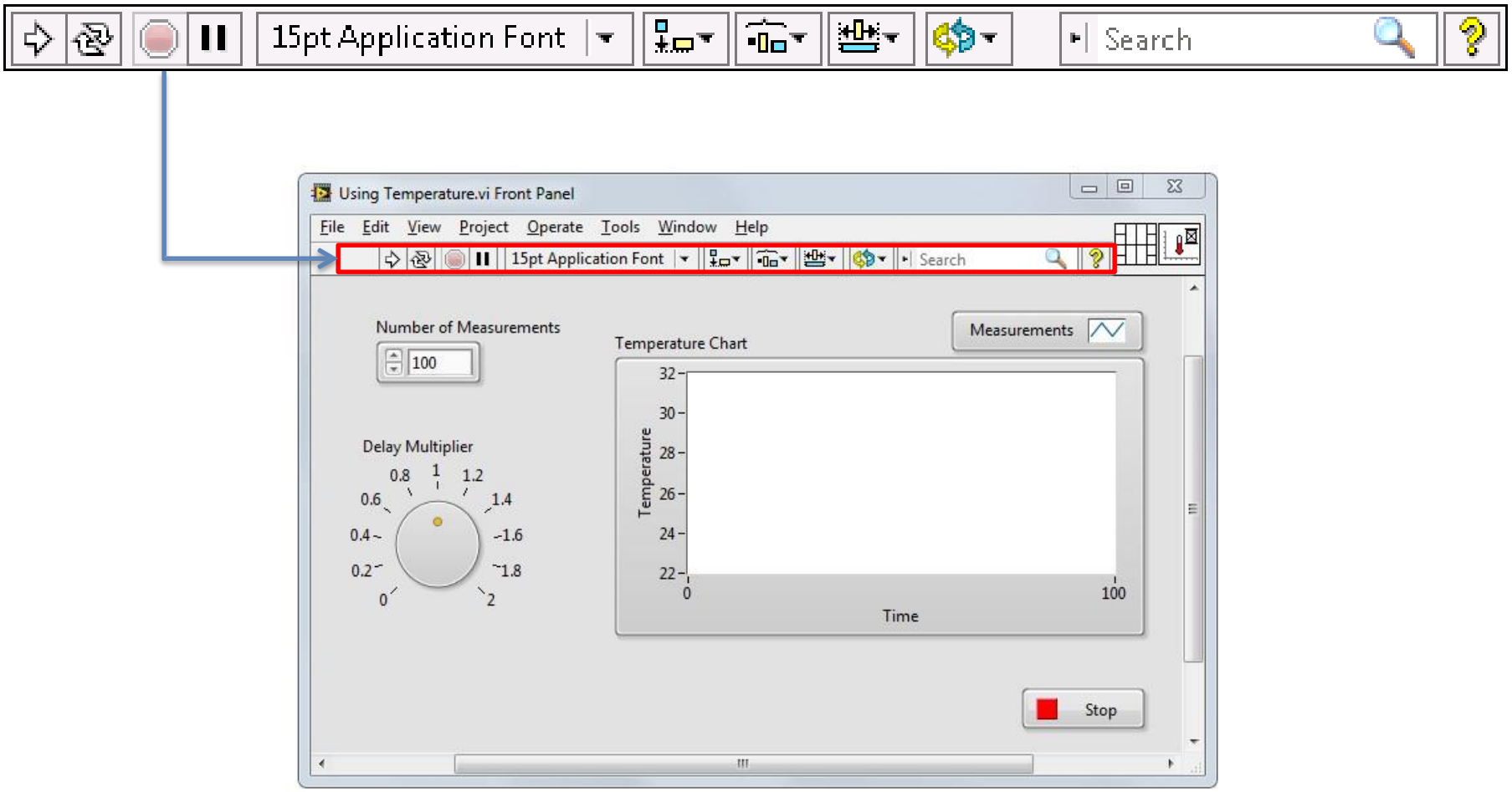
Controls and Indicators

Object Styles

Object Types

- Boolean
- Numeric
- String

Front Panel


15pt Application Font
Search
Using Temperature.vi Front Panel
File Edit View Project Operate Tools Window Help
15pt Application Font
Search
Number of Measurements
100
Delay Multiplier
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2
Temperature Chart
Measurements
Temperature
32 30 28 26 24 22
0
100
Time
Stop

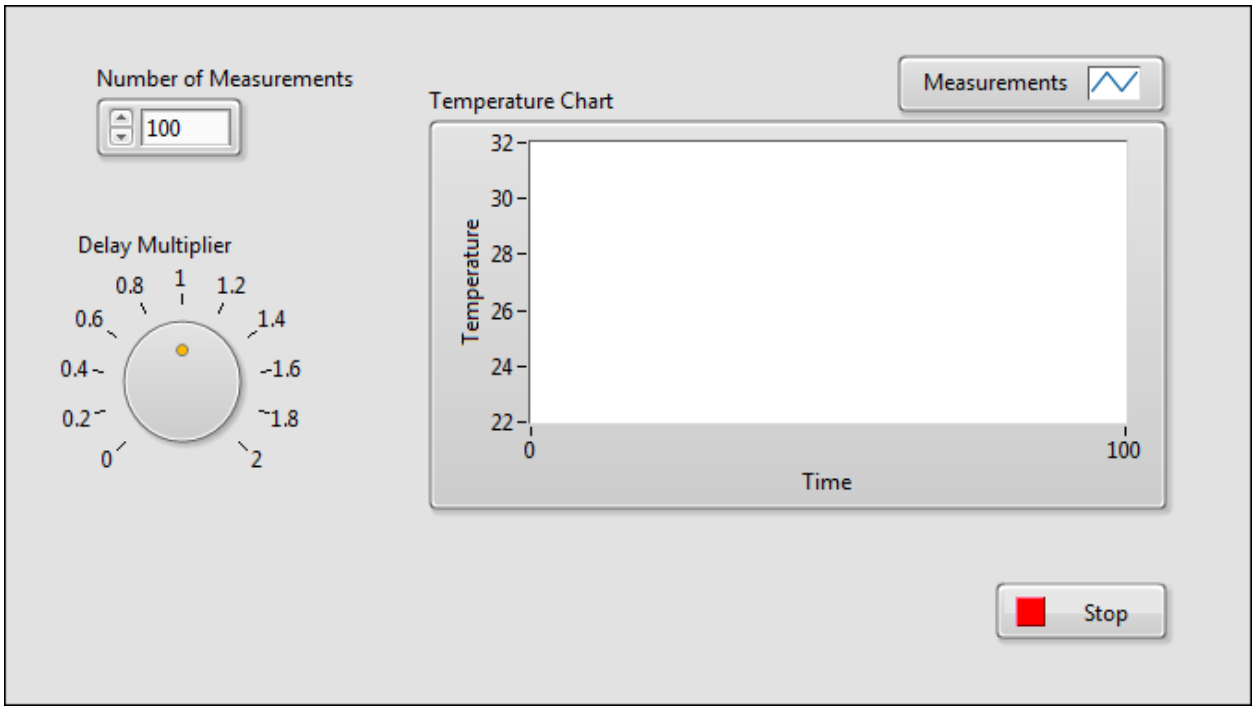
Controls and Indicators

Controls

- Input devices
- Knobs, buttons, slides
- Supply data to the block diagram

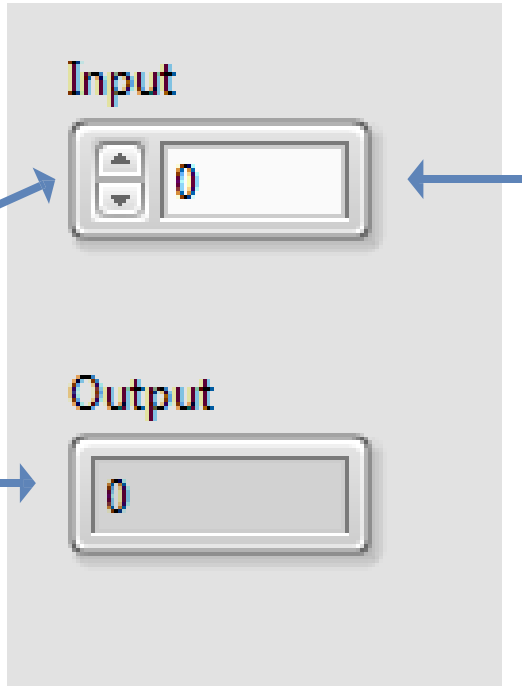
Indicators

- Output devices
- Graphs, LEDs
- Display data the block diagram acquires or generates



Numeric Controls and Indicators

The numeric data in a control or indicator can represent numbers of various types, such as integer or floating-point.



Increment/Decrement buttons

Numeric control

Numeric indicator

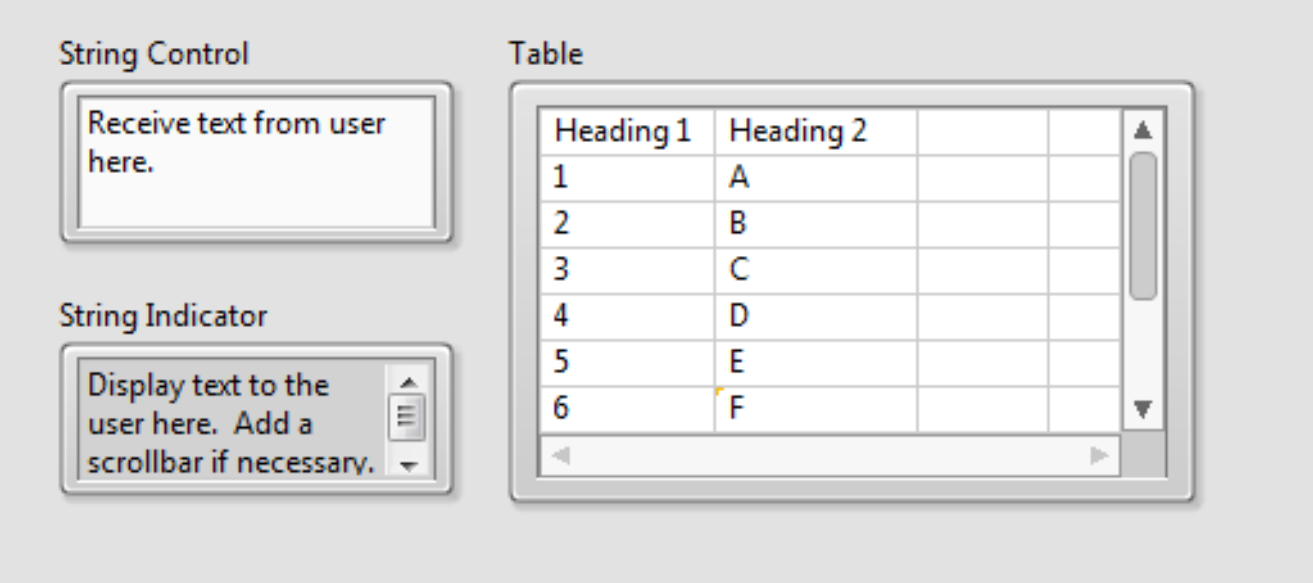
Boolean Controls and Indicators

- The Boolean data type represents data that has only two options, such as True/False or On/Off.
- Use Boolean controls and indicators to enter and display Boolean (TRUE/FALSE) values.
- Boolean objects simulate switches, push buttons and LEDs.



Strings

- The string data type is a sequence of ASCII characters .
- Use string controls to receive text from the user.
- Use string indicators to display text to the user.



E. Block Diagram

Terminals

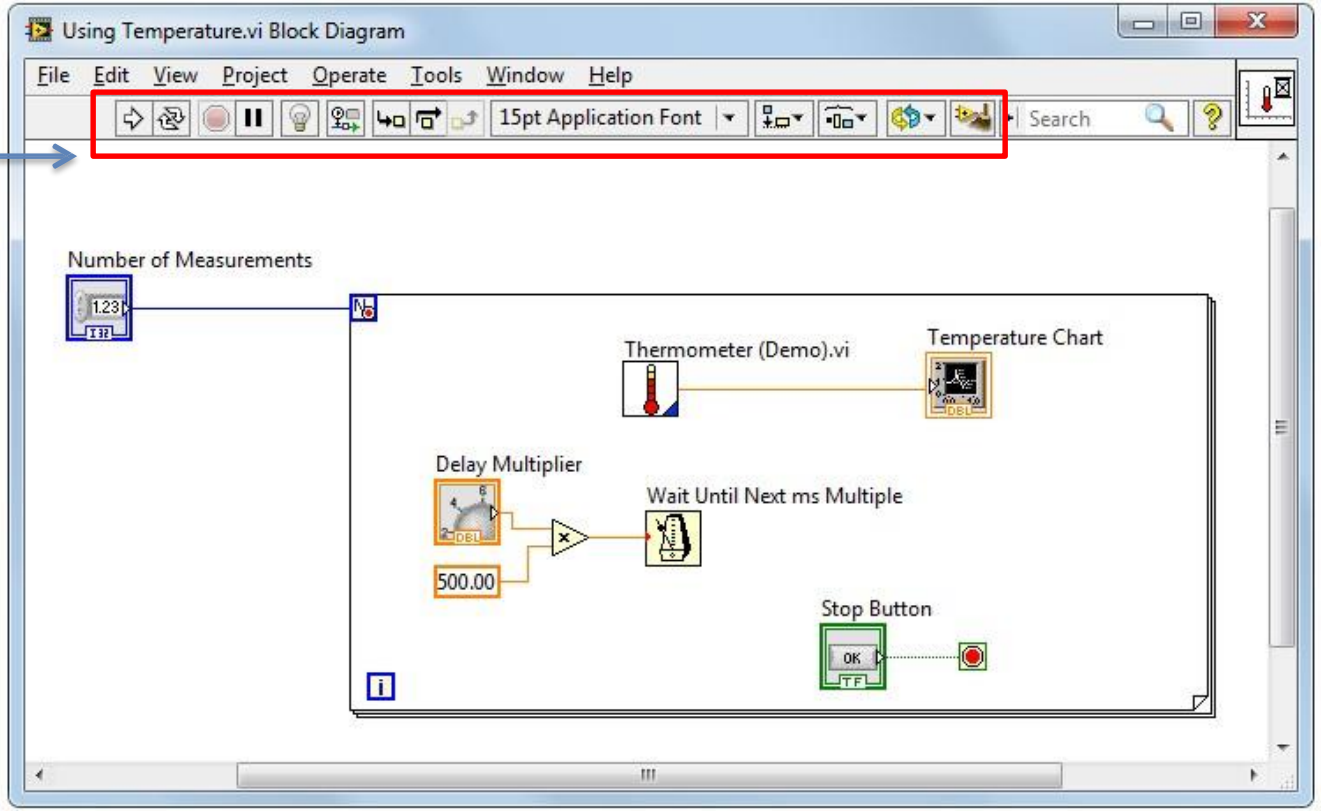
Nodes

Wires

Help

Block Diagram



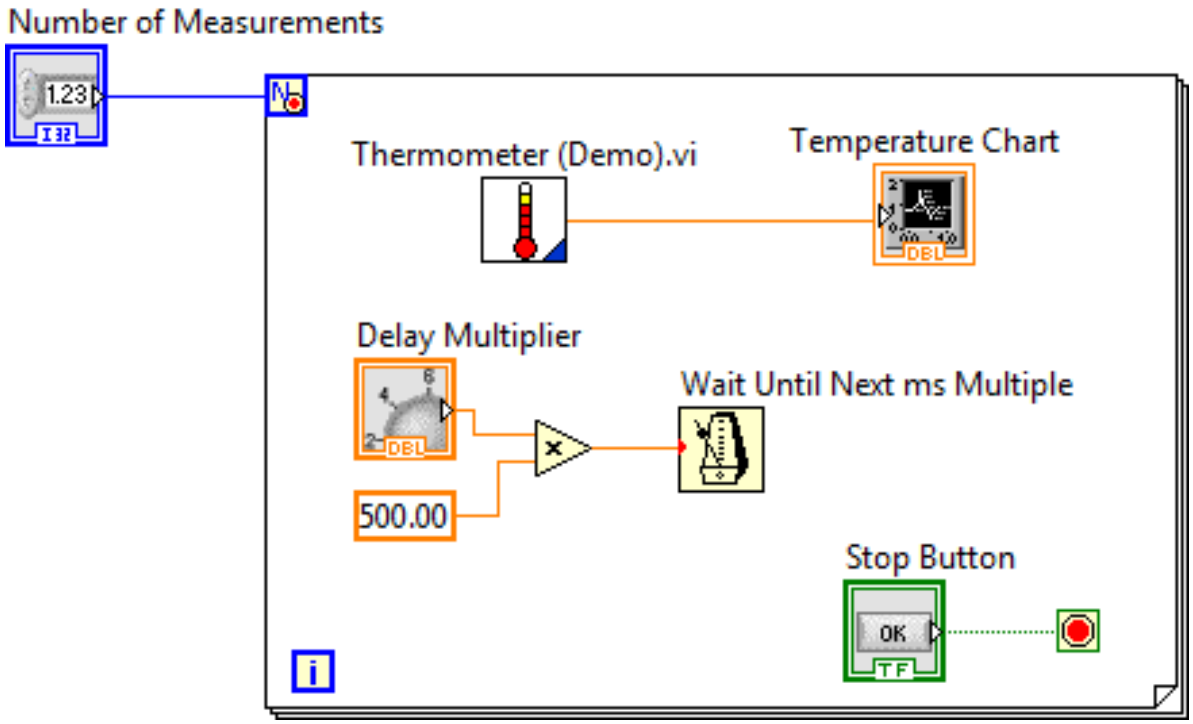


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Block Diagram

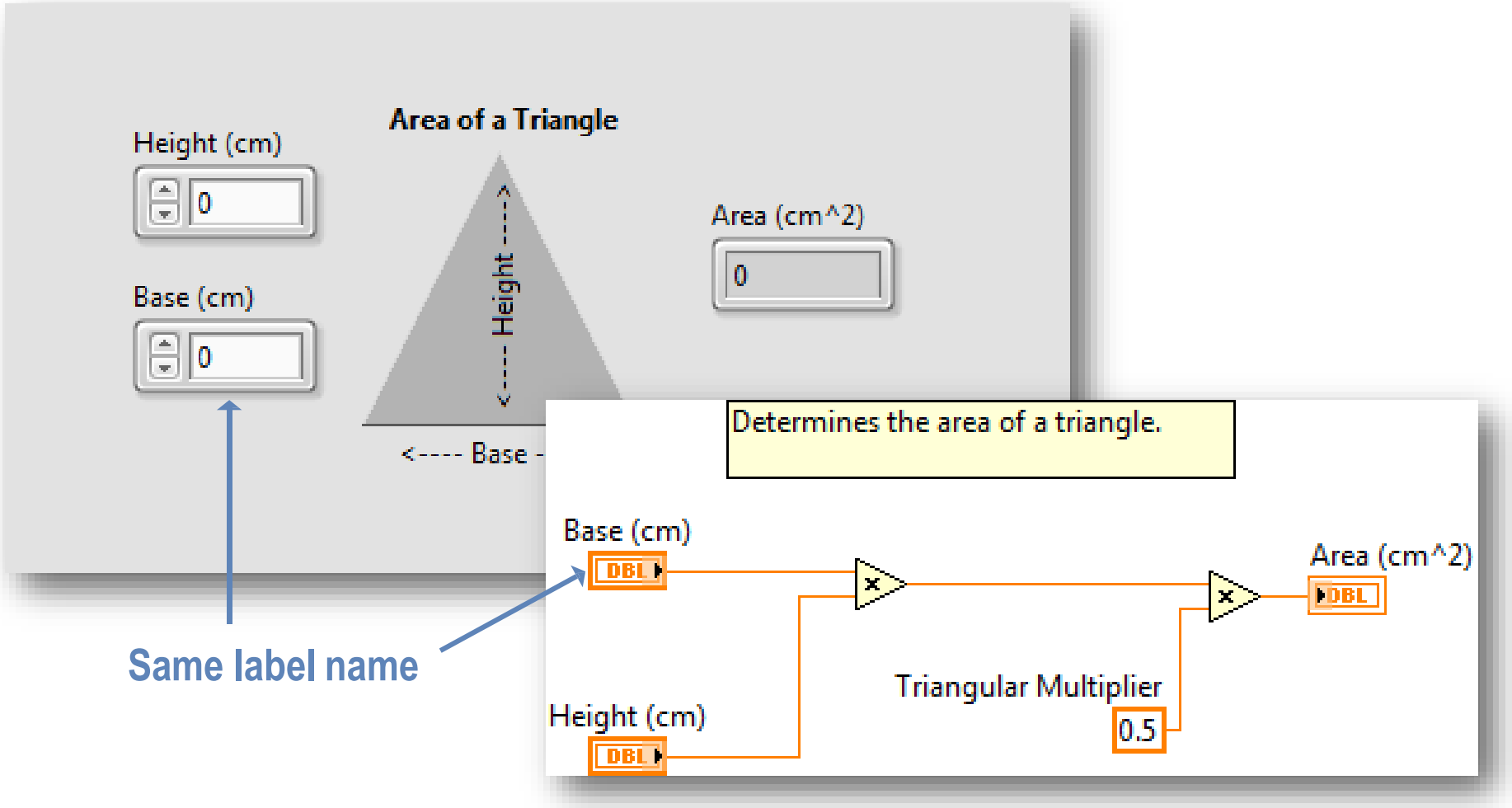
Block diagram items:

- Terminals
- Constants
- Nodes
  - Functions
  - SubVIs
  - Structures
- Wires
- Free labels



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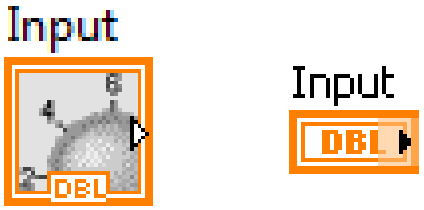
Terminals



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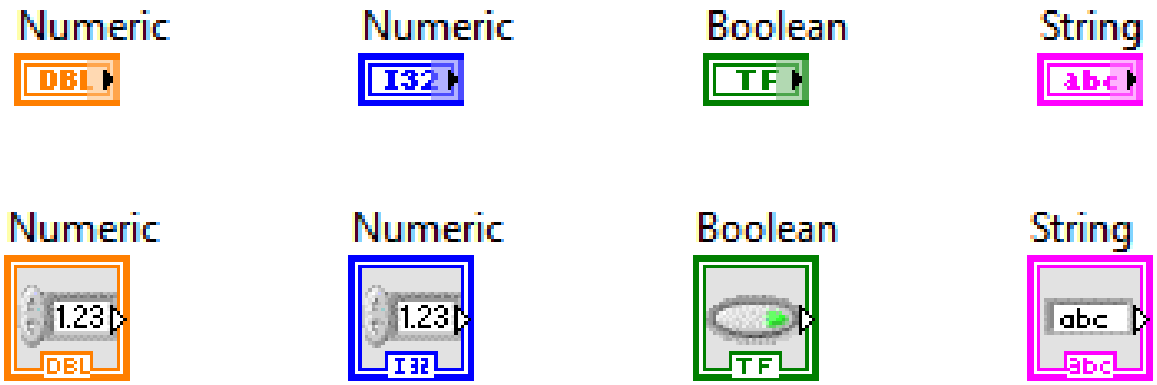
Terminals for Front Panel Objects

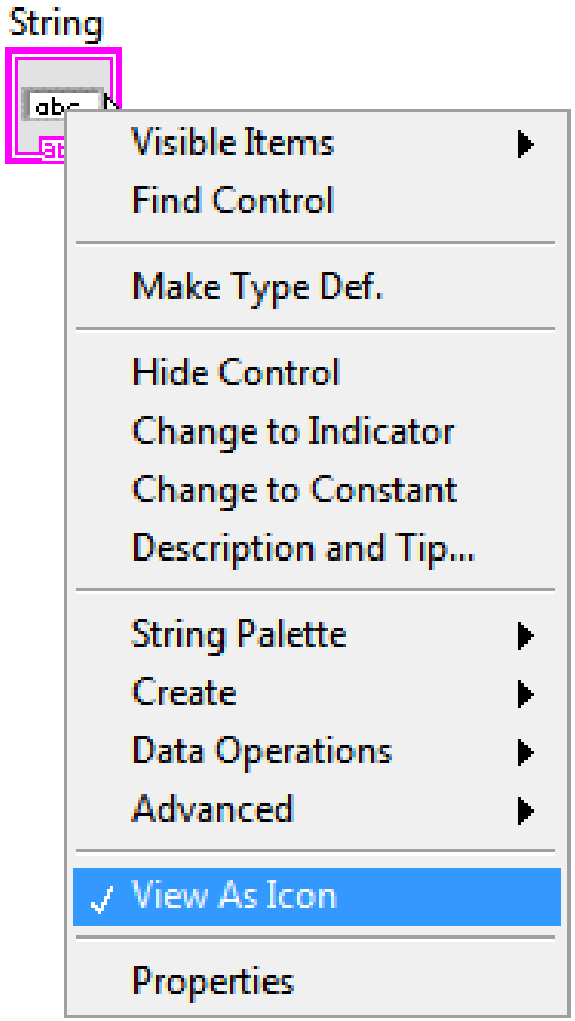
- Terminals are:
  - Entry and exit ports that exchange information between the front panel and block diagram.
  - Analogous to parameters in text-based programming languages.
- Double-click a terminal to locate the corresponding front panel object.



View Terminals as Icons

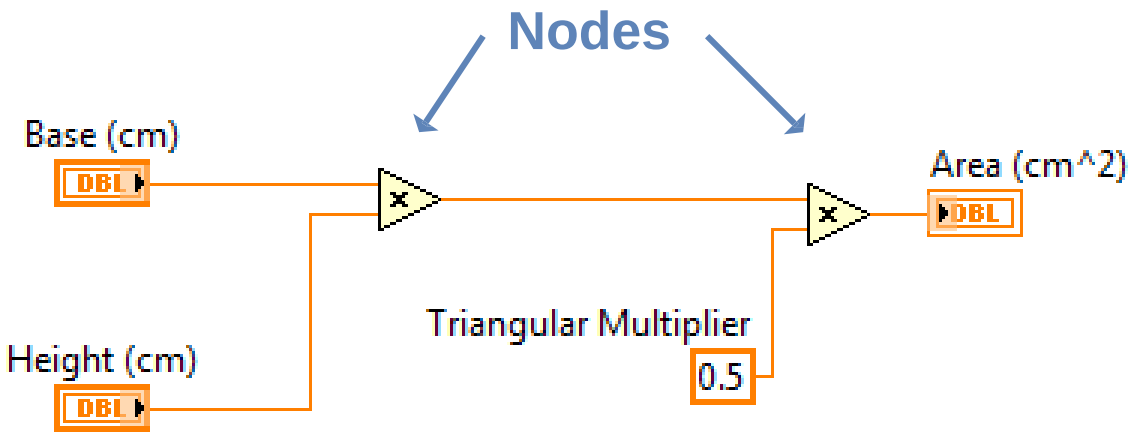
- By default, **View as Icon** option enabled.
- Deselect **View as Icon** for a more compact view.



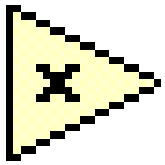


Nodes

Nodes are objects on the block diagram that have inputs and/or outputs and perform operations when a VI runs.

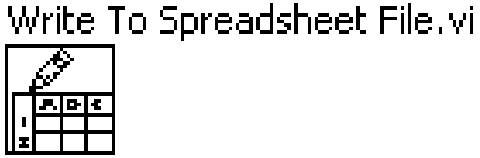


Function Nodes



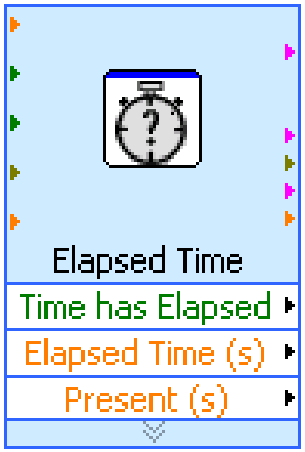
- Functions are:
  - Fundamental operating elements of LabVIEW.
  - Do not have front panels or block diagrams, but do have connector panes.
  - Has a pale yellow background on its icon.
- Double-clicking a function only selects the function.
- Functions do not open like VIs and subVIs.

SubVI Nodes



- SubVIs :
  - Are VIs that you use on the block diagram of another VI.
  - Have front panels and block diagrams.
  - Use the icon from the upper-right corner of the front panel as the icon that appears when you place the subVI on a block diagram.
- When you double-click a subVI, the front panel and block diagram open.
- Any VI has the potential to be used as a subVI.

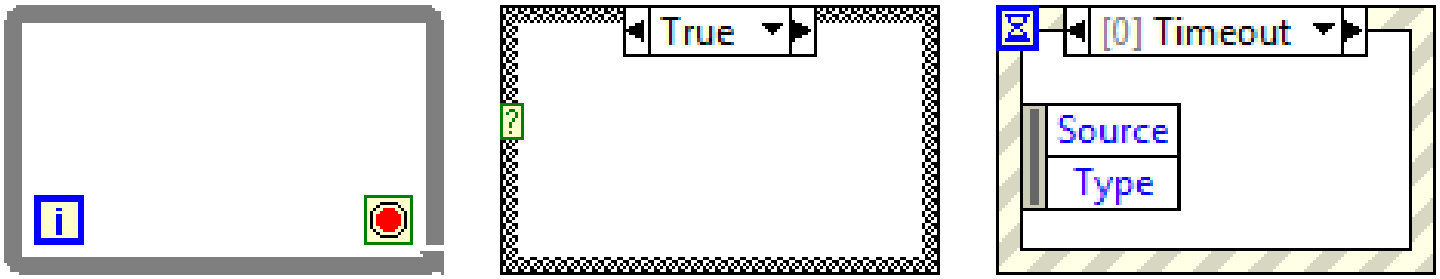
Express VIs



- Express VIs:
  - Are a special type of subVI.
  - Require minimal wiring because you configure them with dialog boxes.
  - Save each configuration as a subVI.
- Icons for Express VIs appear on the block diagram as icons surrounded by a blue field.

Structures

- Structures in LabVIEW have the form of frames.



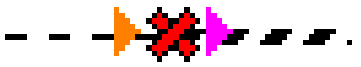
- Other nodes (functions, subVIs, more structures) can be inserted into the frames.

Wires

- Wires transfer data between block diagram objects.
- Wires are different colors, styles, and thicknesses, depending on their data types.

	Floating-point	Integer	String	Boolean
Scalar				
1-D Array				
2-D Array				

- A broken wire appears as a dashed black line with a red X in the middle.



Constants

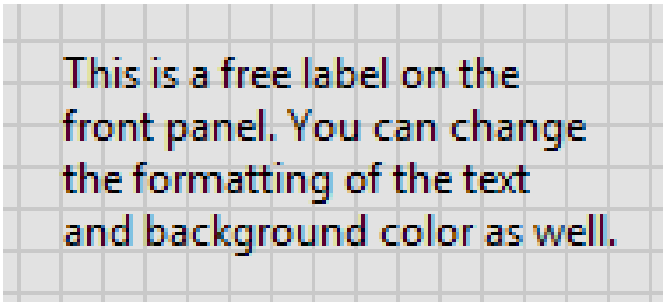
- Constants are the source of values just as control terminals, but their value is fixed in the code.

- You can create a constant of each data type.



Free labels

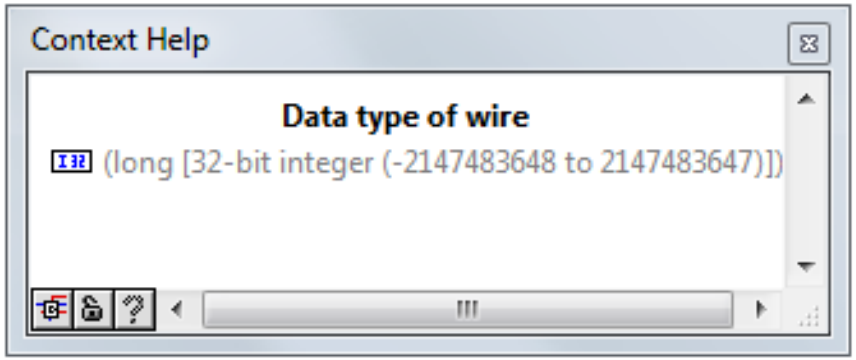
- A free label is a label (a text box) not attached to any object.
- Free labels can be put on the front panel or block diagram. They are created by double-clicking on empty space in the window.
- They can serve as comments or instructions to the user of the application.

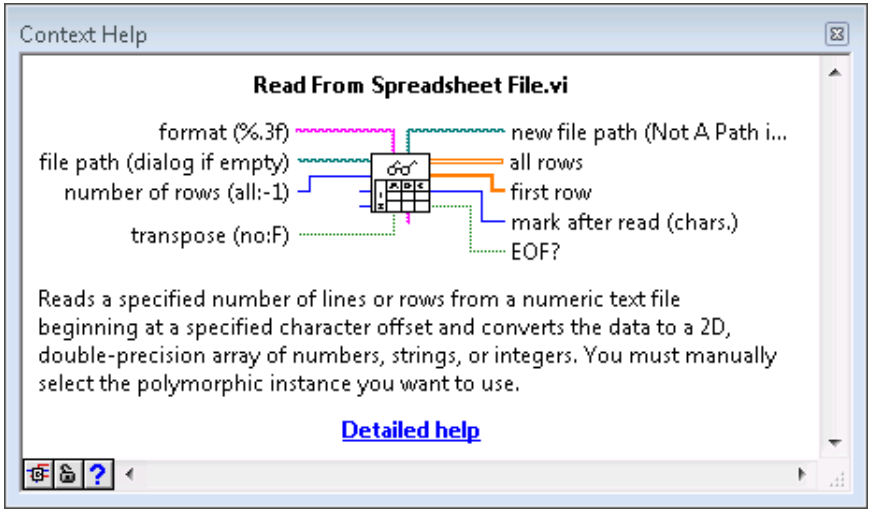


This is the default formatting of a block diagram free label.

Context Help

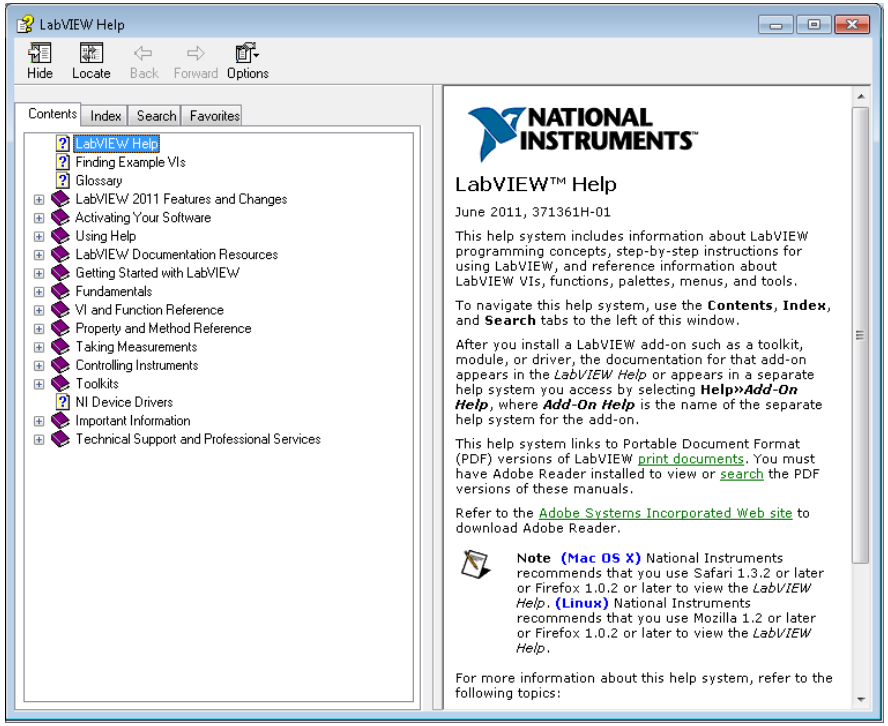
- Displays basic information about wires and nodes when you move the cursor over an object.
- Can be shown or hidden in the following ways:
  - Select **Help»Show Context Help** from the LabVIEW menu.
  - Press <Ctrl-H>.
  - Click the following button on the toolbar: 





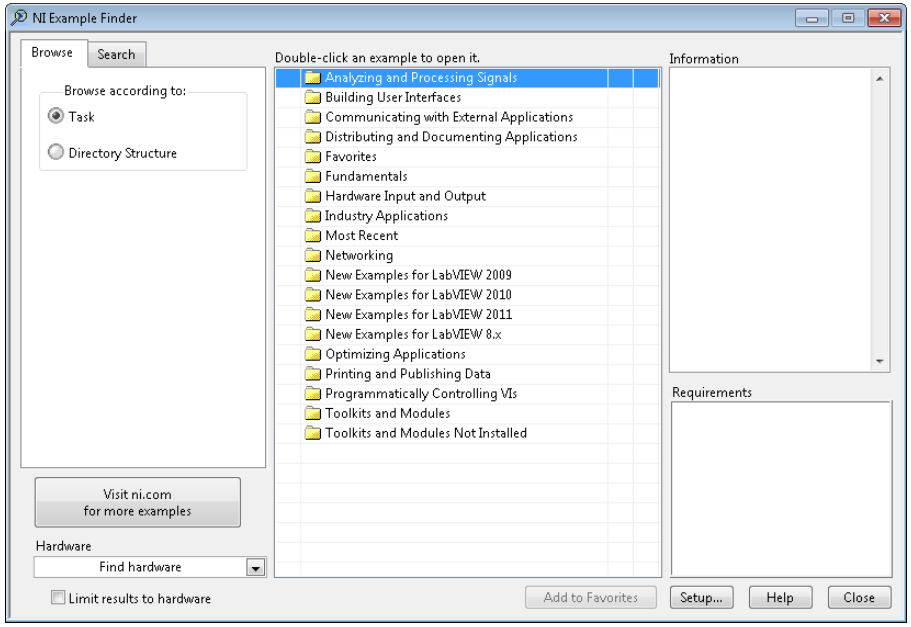
LabVIEW Help

- Contains detailed descriptions and instructions for most palettes, menus, tools, VIs, and functions.
- Can be accessed by:
  - Selecting **Help» LabVIEW Help** from the menu.
  - Clicking the **Detailed help** link in the **Context Help** window.
  - Right-clicking an object and selecting **Help** from the shortcut menu.



Examples

- LabVIEW includes hundreds of example VIs.
- Use NI Example Finder to browse and search installed examples.
  - Select **Help»Find Examples** in the menu.



- Click the example buttons in *LabVIEW Help* topics.



F. Searching for Controls, VIs and Functions

Palettes

Quick Drop

NI Global Search

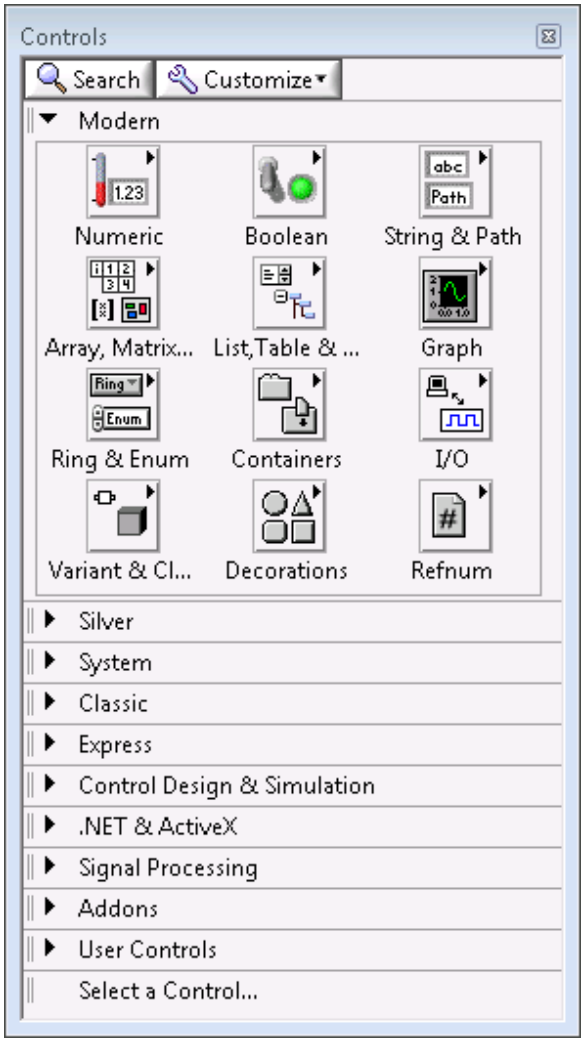
Searching for Controls, VIs and Functions

Ways to find controls, VIs, and functions:

- Search or navigate the palettes.
  - Controls palette
  - Functions palette
- Search by name of object.
  - Quick Drop dialog box
- Search palettes, *LabVIEW Help*, and `ni.com`.
  - Search text box in toolbar

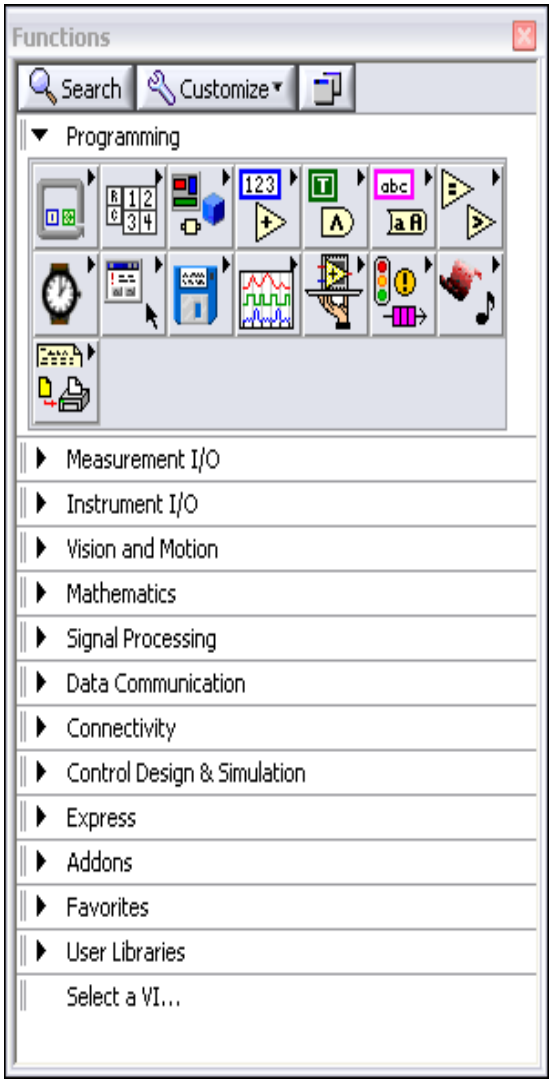
Controls Palette

- Contains the controls and indicators you use to create the front panel.
- Navigate the subpalettes or use the **Search** button to search the Controls palette.



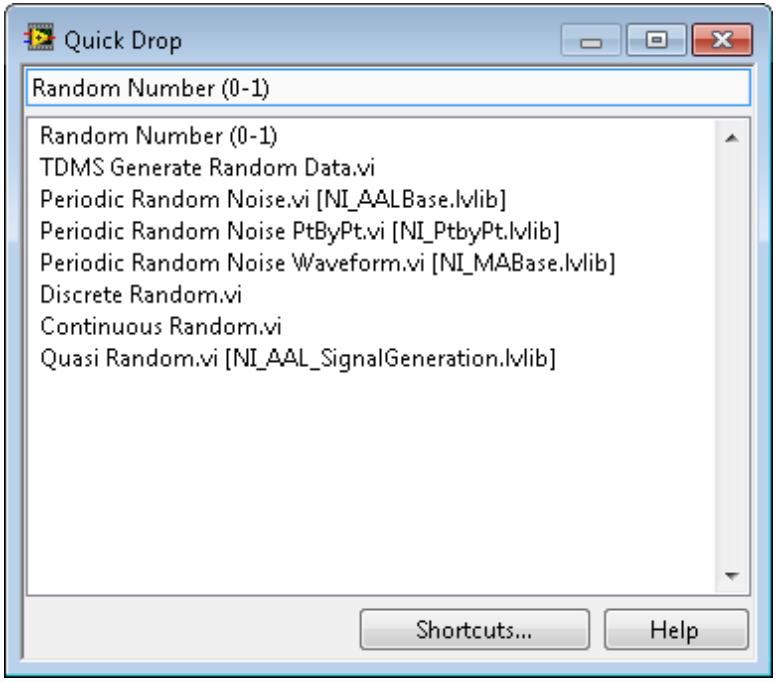
Functions Palette

- Contains the VIs, functions, and constants you use to create the block diagram.
- Navigate the subpalettes or use the **Search** button to search the Functions palette.



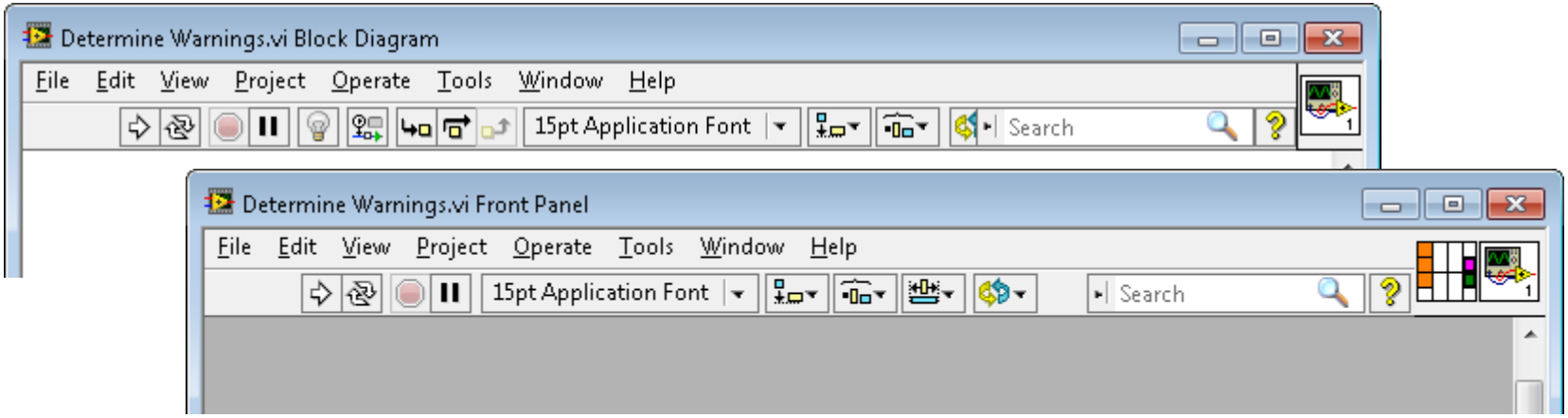
Searching with Quick Drop

- Lets you quickly find controls, functions, VIs, and other items by name.
- Press the <Ctrl-Space> keys to display the Quick Drop dialog box.



Global Search

Use the Search bar in the top right of the front panel and block diagram windows to search palettes, *LabVIEW Help*, and `ni.com`.



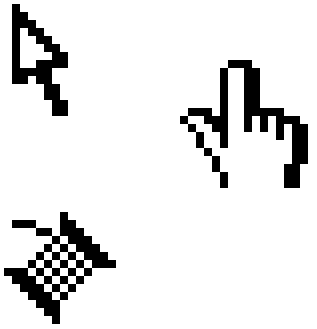
G. Selecting a Tool

Selecting a Tool

Block Diagram Clean-Up

Selecting a Tool

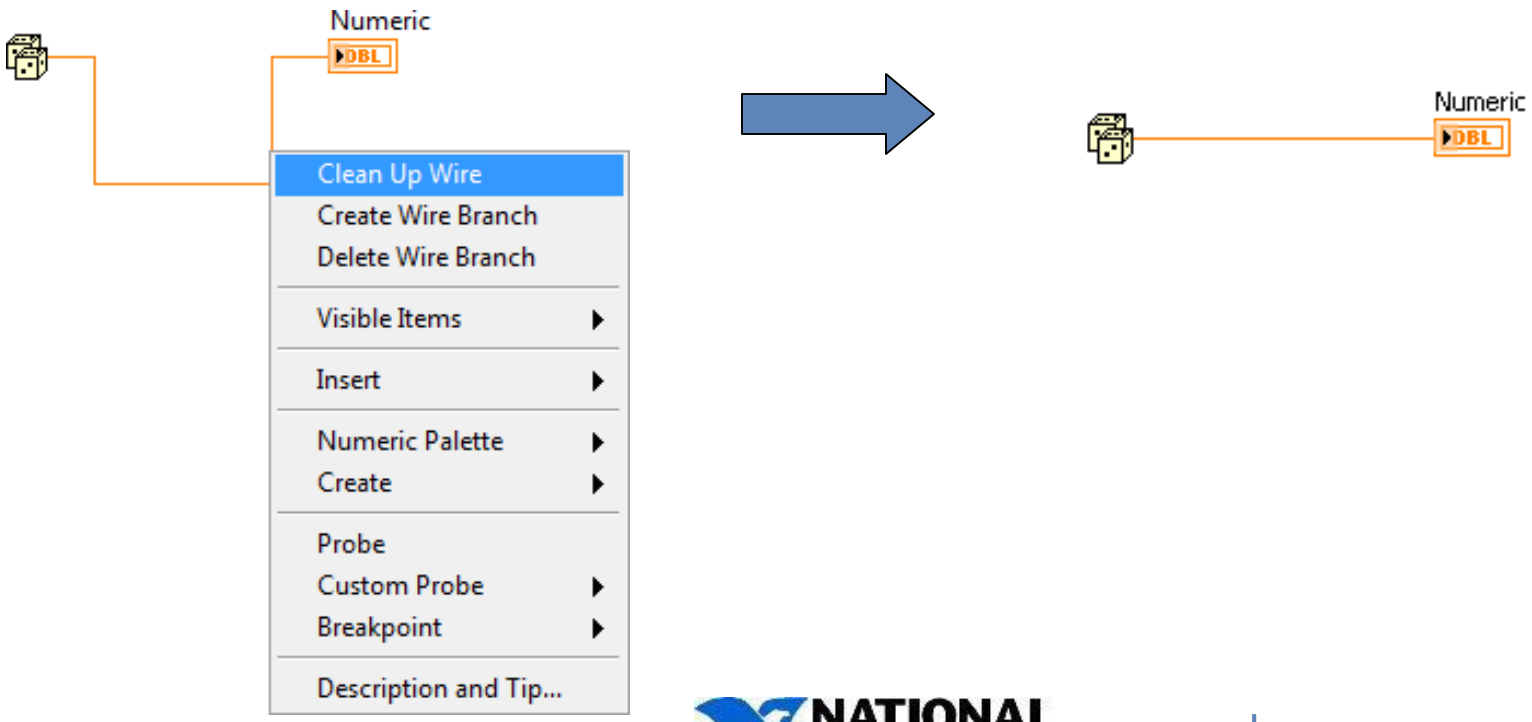
- A tool is a special operating mode of the mouse cursor.
- Create, modify, and debug VIs using the tools provided by LabVIEW.
- By default, LabVIEW automatically selects tools based on the context of the cursor.
- If you need more control, use the **Tools** palette to select a specific tool.
  - Select **View»Tools Palette** to open the **Tools** palette.





Wiring Tips

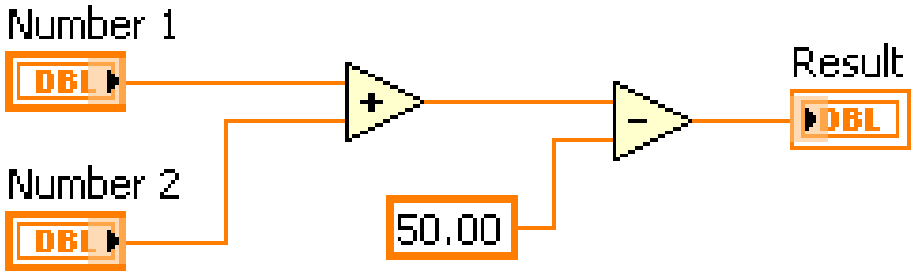
- Press <Ctrl-B> to delete broken wires.
- Right-click and select **Clean Up Wire** to reroute the wire.



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H. Dataflow

Dataflow



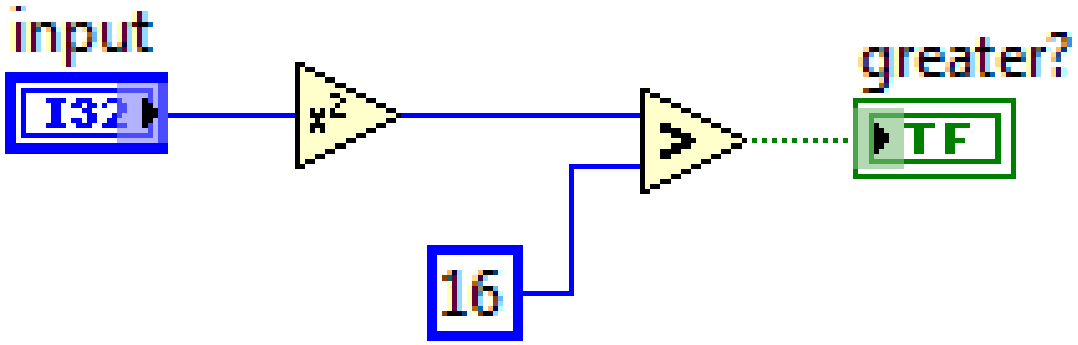
LabVIEW follows a dataflow model for running VIs.

- A node executes only when data are available at all of its required input terminals.
- A node supplies data to the output terminals only when the node finishes execution.

Dataflow – Quiz

What are the nodes in this fragment of code?

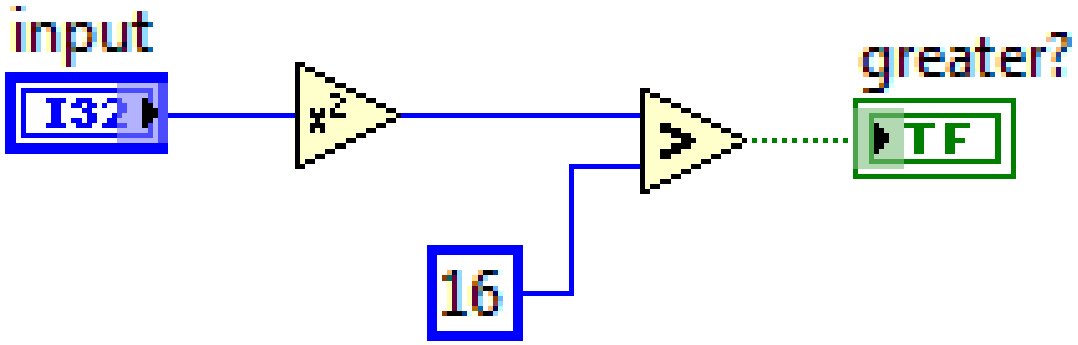
Which node executes first?



Dataflow – Quiz Answer

There are two nodes: „square” and „greater than?” functions.

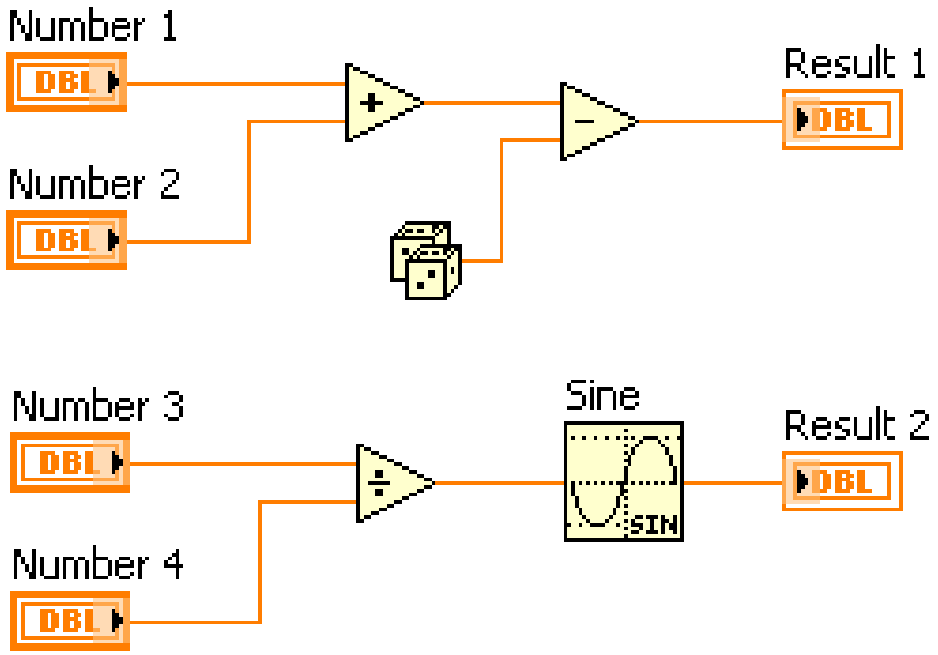
Square executes first.



Dataflow – Quiz

Which node executes first?

a) Add
b) Subtract
c) Random Number
d) Divide
e) Sine



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Dataflow – Quiz Answer

No single correct answer.

Which node executes first?

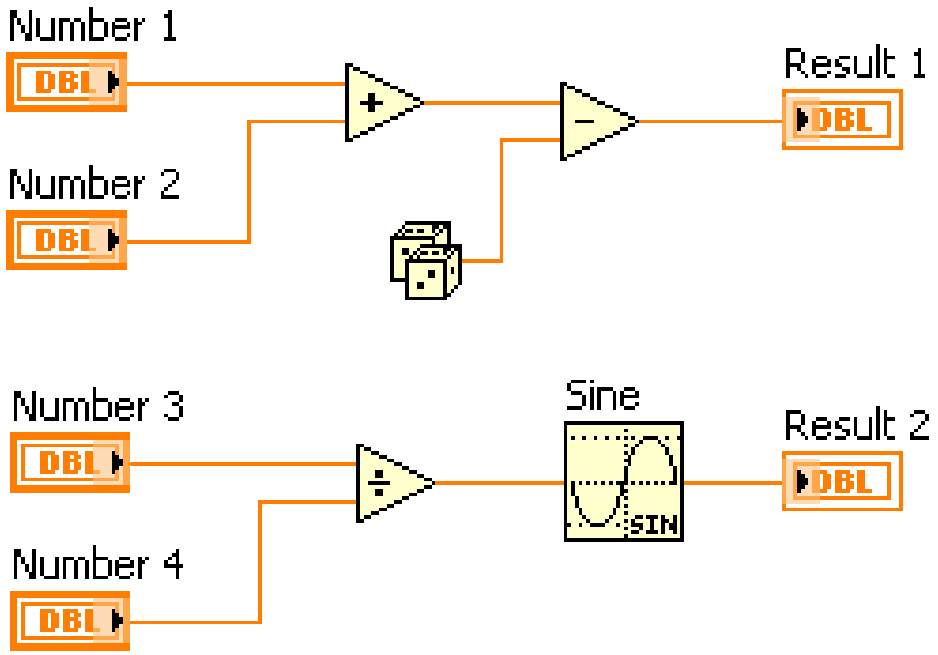
a) Add – **Possibly**

b) Subtract – **Definitely not**

c) Random Number – **Possibly**

d) Divide – **Possibly**

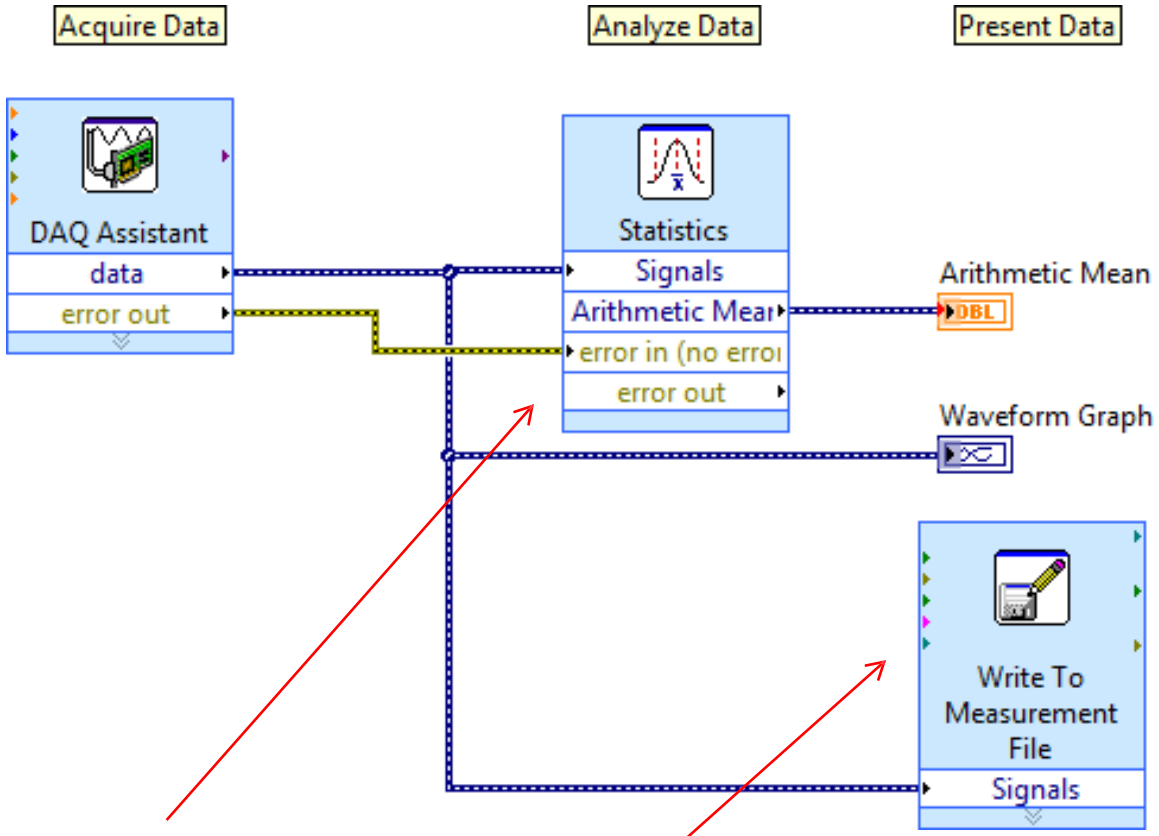
e) Sine – **Definitely not**



Group Exercise
Concept: Dataflow

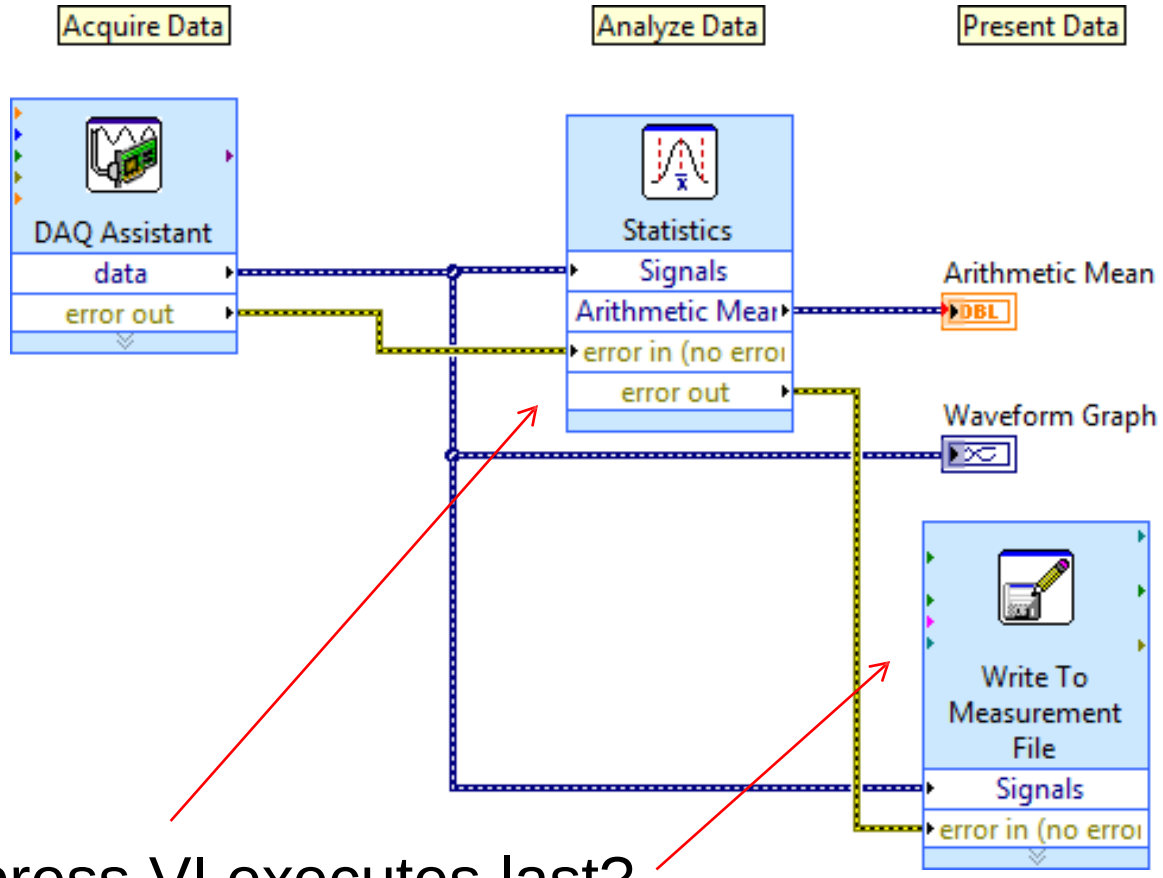
Identify dataflow execution order in the following block diagrams.

Group Exercise
Concept: Dataflow



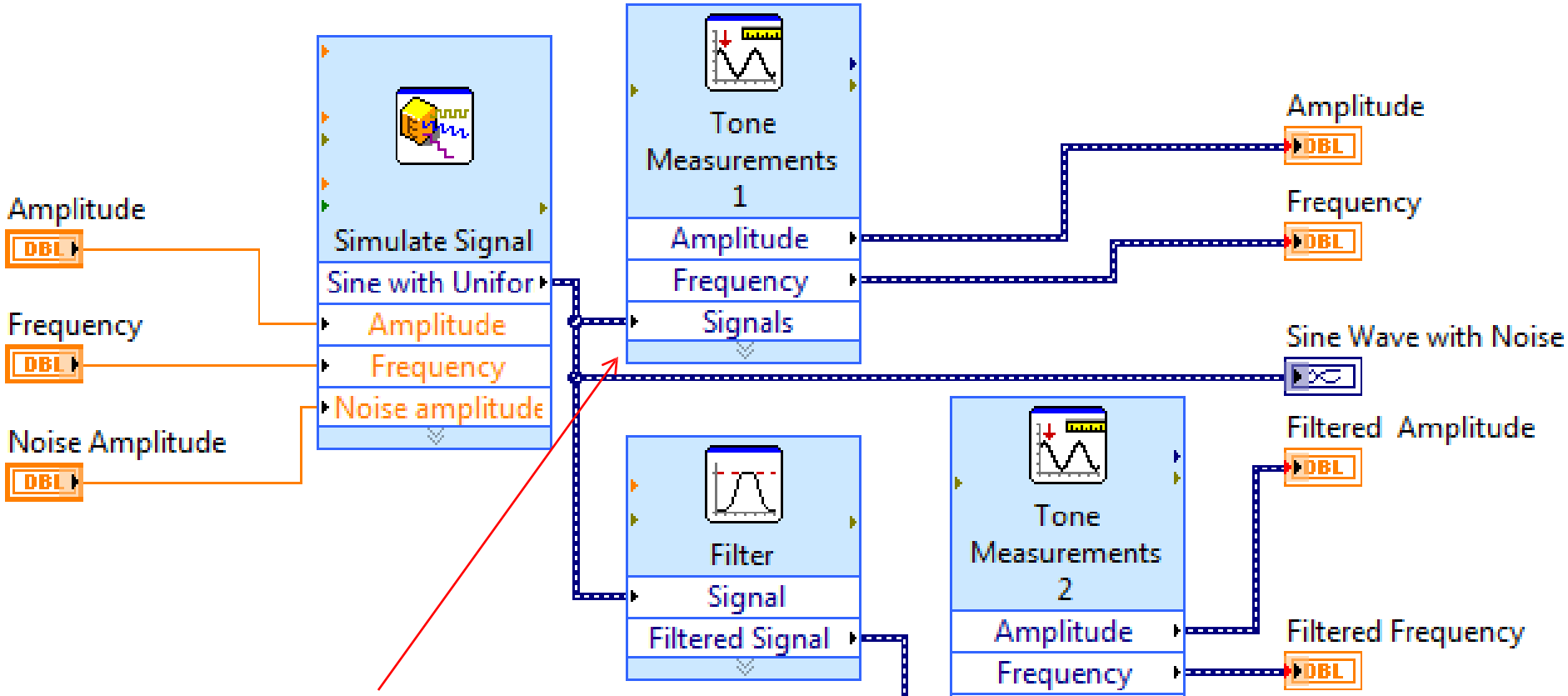
Which Express VI executes last?

Group Exercise
Concept: Dataflow



Which Express VI executes last?

Group Exercise
Concept: Dataflow



Which Tone Measurements Express VI executes last?


1.23
Ring
Enum